

VANCOUVER PEARSON, WA, USA

WMO: 727918

Lat: 45.621N

Lon: 122.657W

Elev: 30

StdP: 14.68

Time Zone: -8.00 (NAP)

Period: 98-19

WBAN: 94298

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	24.4	27.4	9.0	8.7	31.9	16.2	12.5	33.5	20.6	41.0	18.3	42.5	2.7	140	0.469

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	22.9	91.4	66.7	87.7	66.1	83.6	64.7	69.1	86.5	67.5	84.3	65.9	81.3	8.7	310

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
62.9	85.8	74.6	61.2	80.7	72.8	59.5	76.0	71.6	33.1	86.3	31.7	83.9	30.5	81.6	77.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min	Max		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.3	13.1	11.7	18.2	99.1	5.0	3.2	14.6	101.4	11.7	103.2	8.9	105.0	5.2	107.3
			17.0	72.0	5.0	2.1	13.4	73.5	10.4	74.8	7.6	76.0	4.0	77.5
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	54.2	41.0	43.2	47.4	51.6	58.1	63.4	69.1	69.6	64.0	54.7	46.6	40.8
	DBStd	11.57	6.08	5.09	4.84	5.18	5.85	5.68	4.95	4.78	5.28	4.91	6.13	6.30
	HDD50	1069	284	195	110	38	4	0	0	0	0	14	133	291
	HDD65	4359	745	611	547	402	228	95	15	12	81	320	552	751
	CDD50	2601	4	4	28	88	255	403	592	609	421	161	31	5
	CDD65	413	0	0	0	1	14	48	142	155	52	1	0	0
	CDH74	4547	0	0	0	49	278	602	1440	1508	648	22	0	0
	CDH80	1728	0	0	0	9	88	230	562	615	222	2	0	0
Wind	WSAvg	4.6	4.8	4.9	4.8	4.5	4.6	4.8	5.3	4.7	4.0	3.7	4.5	5.1
Precipitation	PrecAvg	40.8	6.1	4.4	4.1	3.1	2.5	1.7	0.7	1.1	1.8	3.3	6.1	6.7
	PrecMax	64.4	12.8	10.6	7.2	7.7	6.2	4.0	3.7	5.1	4.8	7.4	12.9	13.3
	PrecMin	27.9	0.3	0.8	0.9	0.9	0.2	0.0	0.0	0.0	0.0	0.2	1.1	1.7
	PrecStd	7.6	2.8	2.2	1.5	1.4	1.3	1.0	0.7	1.1	1.3	1.8	2.9	2.8
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	57.4	60.9	70.0	80.6	88.1	94.1	97.0	96.9	91.7	77.2	64.8	59.2
		MCWB	52.9	48.7	54.8	60.6	63.4	67.6	68.8	68.4	65.6	59.7	57.7	55.6
	2%	DB	54.5	56.8	64.2	72.7	81.7	86.3	90.6	91.2	86.2	71.7	61.0	54.5
		MCWB	50.1	48.2	51.8	56.4	61.1	65.6	67.2	66.9	63.4	57.3	55.6	50.9
	5%	DB	52.3	54.3	60.2	66.4	75.4	81.1	86.1	86.9	81.6	67.6	57.7	51.9
		MCWB	48.4	47.3	50.1	53.1	59.1	63.3	66.5	66.5	63.0	56.4	52.9	48.5
	10%	DB	49.7	51.9	56.5	62.5	70.5	75.4	81.9	82.2	76.8	63.7	55.2	49.0
		MCWB	46.1	46.2	49.0	52.0	57.3	60.9	65.0	65.2	61.6	55.6	50.9	45.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	55.0	53.1	57.0	61.9	66.0	70.0	71.9	71.1	69.0	63.6	60.1	55.9
		MCDB	57.1	56.1	66.3	77.6	84.8	89.8	90.0	91.0	85.2	71.3	62.9	59.0
	2%	WB	51.1	50.7	53.8	57.7	62.8	66.5	68.9	69.1	66.2	60.4	56.2	51.7
		MCDB	53.4	54.6	60.0	69.5	78.2	83.3	85.8	86.6	80.6	67.1	59.8	54.1
	5%	WB	48.5	48.5	51.7	55.0	60.4	64.1	67.2	67.4	64.3	58.6	53.9	48.8
		MCDB	51.4	52.5	57.6	64.1	72.9	78.3	84.2	83.8	77.3	64.4	56.9	51.1
	10%	WB	46.4	46.6	50.1	52.8	58.3	61.9	65.5	65.7	62.5	56.9	52.0	46.2
		MCDB	48.9	50.3	55.9	60.0	68.5	73.9	81.0	80.7	74.0	62.2	54.9	48.4
Mean Daily Temperature Range	5% DB	MDBR	11.2	14.2	15.7	17.3	18.6	19.1	22.3	22.9	22.0	16.9	12.5	10.4
		MCDBR	13.4	18.9	23.8	28.9	31.2	30.5	30.8	31.9	32.0	25.4	14.2	11.8
		MCWBR	9.7	11.6	13.6	14.7	14.4	13.0	12.3	12.2	13.1	13.1	9.3	9.1
	5% WB	MCDBR	12.3	14.1	18.6	24.6	26.4	27.1	27.6	28.0	26.8	17.7	11.0	10.7
		MCWBR	9.7	9.5	11.8	13.5	12.8	12.3	12.0	11.6	11.8	10.4	8.3	8.9
Clear-Sky Solar Irradiance	taub		0.320	0.317	0.330	0.361	0.377	0.362	0.350	0.357	0.343	0.334	0.322	0.326
	taud		2.532	2.553	2.524	2.403	2.375	2.431	2.491	2.486	2.506	2.545	2.538	2.476
	Ebn at Noon		252	277	287	285	283	287	289	283	279	266	248	234
	Edh at Noon		21	24	29	35	37	36	33	32	29	24	21	20
All-Sky Solar Radiation	RadAvg		366	632	920	1341	1707	1886	2146	1833	1369	791	427	299
	RadStd		60	89	121	126	164	177	151	124	120	99	58	45

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+304	+142
Regional (32 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+169	+47

Nomenclature: See separate page